SESSION 2

Variables and Data Types in Python

Objective

Practice declaring variables, checking data types, formatting output, accepting user input, and converting values to calculate a final product price.

Task 1 — Variables and data types

```
name = "Jinal"
age = 22
city = "Ahmedabad"
favorite_food = "Pizza"
has_spotify_subscription = True

print(name, type(name))
print(age, type(age))
print(city, type(city))
print(favorite_food, type(favorite_food))
print(has_spotify_subscription, type(has_spotify_subscription))
```

Example output

```
Jinal <class 'str'>
22 <class 'int'>
Ahmedabad <class 'str'>
Pizza <class 'str'>
True <class 'bool'>
```

Explanation: `str` stores text, `int` stores whole numbers, and `bool` stores `True` or `False`. The `type()` function reports a variable's data type. Replace sample personal values with your own if needed.

Task 2 — Zomato order summary

```
restaurant_name = "Pizza Palace"
item_count = 3
total_price = 450.5
is_veg = True
delivery_time_minutes = 30

print(
    f"Ordered {item_count} items from {restaurant_name}. "
    f"Total: ■{total_price}. Veg: {is_veg}. "
    f"Delivery in {delivery_time_minutes} min."
)
```

Output

```
Ordered 3 items from Pizza Palace. Total: ■450.5. Veg: True. Delivery in 30 min.
```

Explanation: An f-string (a string prefixed with `f`) inserts variable values wherever curly braces `{}` appear.

Task 3 — YouTube channel input

```
followers = int(input("Enter number of followers: "))
```

```
posts = int(input("Enter number of posts: "))

print(f"You have {followers} followers and {posts} posts.")
```

Example run

```
Enter number of followers: 1500
Enter number of posts: 25
You have 1500 followers and 25 posts.
```

Explanation: `input()` receives text from the user. `int()` converts that text to a whole number so it can be stored as an integer. YouTube more commonly uses the terms subscribers and videos, but this example follows the assignment wording.

Task 4 — Convert types and calculate a discount

```
product_price = '499'
discount = 50
is_flash_sale = 'True'

product_price = int(product_price)
is_flash_sale = bool(is_flash_sale)

final_price = product_price - discount

print("Product Price:", product_price)
print("Flash Sale:", is_flash_sale)
print("Discount:", discount)
print("Final Price:", final_price)
```

Output

```
Product Price: 499
Flash Sale: True
Discount: 50
Final Price: 449
```

Explanation: `int('499')` converts the text '499' to integer 499. `bool('True')` returns True because the string is non-empty. The final price is 499 – 50 = ■449.

Important: `bool('False')` *also returns True because 'False' is a non-empty string. To interpret the text correctly, use* `is_flash_sale = is_flash_sale == "True"`.

Quick revision — Python data types

Data type / function	Example	Purpose
String (str)	"Jinal"	Stores text
Integer (int)	22	Stores whole numbers
Float (float)	450.5	Stores decimal numbers
Boolean (bool)	True	Stores True or False
type()	type(age)	Checks a value's data type
input()	input("Enter age: ")	Accepts user input

How to run your practice file

Save the code in VS Code as `variables_practice.py`, then run this command in the terminal:

```
python variables_practice.py
```

Assignment tip: Run each task on your own computer and compare your output with the examples above.